HIGH-LEVEL DESIGN (HDD)

Academic Writing Journey

A retrospective High-Level Design — written as if authored before development began, applying enterprise Learning Experience Architecture standards to an existing, independently evaluated eLearning product.

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> ***Note on this document:*** *The Academic Writing Journey (AWJ) was built and independently evaluated before this HDD was written. This document reconstructs the design architecture retrospectively, in the format and rigour expected of a High-Level Design at the Learning Experience Architect level — demonstrating the ability to think in HDD structure, not only to produce finished learning content.*

1 EXECUTIVE SUMMARY

Academic Writing Journey (AWJ) is a fully self-paced, gamified eLearning solution designed to close a specific performance gap: university-level learners in English for Specific/Academic Purposes (ESP/EAP) consistently struggle to produce a structurally sound, publishable academic abstract and to deliver a confident spoken conference pitch. The solution architecture spans six progressive levels, a bonus escape-room module, AI-assisted authorship support, and a full accessibility layer aligned to WCAG and the EU Accessibility Act (EAA).

Independent evaluation by Universitat Politècnica de València (UPV, 2026) rated the solution 10/10 for inclusive eLearning design — validating both the instructional architecture and the accessibility strategy described in Section 6.

2 BUSINESS CONTEXT & PERFORMANCE GAP

2.1 Stakeholder & Sponsor

Department of Applied Linguistics, Universitat Politècnica de València — course: "Simulation, Games and Technology Applied to Teaching Strategies and Research Methodology," Prof. Casañ Pitarch, Ricardo.

2.2 Performance Gap Identified

- Learners enter academic writing tasks with strong general English proficiency (B2) but **lack the genre-specific rhetorical structure** required for academic abstracts (Swales' Move Analysis) and the metadiscourse competence needed for persuasive academic register (Hyland's framework).
- Existing static course materials produced passive recognition of academic structure but **did not transfer into independent production** — a classic input-without-output gap (Swain's Output Hypothesis).
- No existing institutional solution combined rigorous SLA theory with sustained learner engagement across a fully asynchronous, self-paced format — engagement, not content availability, was the primary risk to program success.

2.3 Target Audience

Undergraduate (Years 2–4) and Master's-level (Years 1–2) students requiring B2→C1 academic English for research writing and conference presentation, across mixed in-person, online, and hybrid modalities.

3 SOLUTION ARCHITECTURE

3.1 Instructional Model

- **SAM** (Successive Approximation Model): iterative analyse–prototype–review cycles replacing linear waterfall development, allowing each level to be pedagogically refined against real learner UX testing.
- **IPO Model** (Casañ-Pitarch): six levels mapped explicitly to Input (L1–2) → Processing (L3–4) → Output (L5–6), ensuring a structured progression from receptive to productive skill.
- ESA + CLIL 4Cs: Engage–Study–Activate combined with Content, Communication, Cognition, and Culture within authentic academic contexts.

3.2 Modality Approach

Modality	Role in Solution	Delivery Tooling
Self-paced digital	Core delivery — all 6 levels + bonus escape room, fully asynchronous	HTML5 / Tailwind CSS / GitHub Pages
AI-assisted tutoring	On-demand scaffolding via conversational AI tutor for in-the-moment learner support	“PhD Chema GPT” (custom GPT)
Asynchronous social	Peer feedback and community reinforcement without synchronous scheduling	Padlet, Flip
Multimedia / video	Worked examples and AI-avatar instructional delivery for key concepts	Synthesia, YouTube/TED-Ed

3.3 Gamification & Engagement Architecture

- **Progressive levelling**: 6 sequential levels, each gated on completion of the prior, preventing cognitive overload while maintaining narrative momentum.
- **XP economy**: 100–1,000 XP per level (2,500 XP total) plus a 180 XP bonus escape-room module — calibrated reward density to sustain motivation across a multi-week, fully self-paced journey.
- Narrative identity arc: learner progresses from “Curious Student” to “Academic Challenger” (boss level), applying **Self-Determination Theory** (autonomy, competence, relatedness) to an asynchronous format that cannot rely on instructor-driven motivation.
- Boss-level gate: Level 6 (Conference Pitch) is unlocked only via the bonus escape room — a deliberate friction point ensuring mastery of Levels 1–4 metadiscourse content before the highest-stakes production task.

4 AI INTEGRATION & RESPONSIBLE USE

AI was used as a documented technical co-author (Claude Code, Anthropic) for implementation — never as a substitute for pedagogical decision-making, which remained the sole responsibility of the Subject Matter Expert. This distinction was formalised as an explicit governance principle, not an informal practice:

Decision Domain	Owner
Pedagogical sequencing, SLA framework selection, assessment criteria	Human (Subject Matter Expert)
HTML/CSS/JS implementation, UX build, accessibility layer coding	AI (Claude Code) — under human direction and review

This model anticipates the kind of responsible AI-enabled design practice required for enterprise learning architecture: AI accelerates production and exploration of solution options, while domain expertise retains final authority over learner-facing decisions.

5 MEASUREMENT & EVALUATION APPROACH

- **Independent third-party evaluation**: UPV (2026) scored the solution against four pillars — technological/content accessibility, flexibility/personalisation, interaction/collaboration, and digital competencies/AI integration — achieving a 9.0+ average and 10/10 overall for inclusive design.
- UX testing signal: **zero learner navigation errors** recorded during structured usability testing, validating the IPO-aligned information architecture and Cognitive Load Theory (Sweller) chunking strategy.
- Built-in iteration hooks: the evaluator’s documented “areas for improvement” (e.g. anonymous leaderboard option, pre-Level-1 tool tutorial, formal WCAG audit) were captured as a structured backlog for the next design iteration — modelling a pilot-and-refine mindset rather than a one-time build.

6 ACCESSIBILITY, INCLUSION & GOVERNANCE

Accessibility was treated as a core design constraint from the outset, not a post-launch remediation:

- A floating **Universal Design for Learning** (UDL) control layer — voice dictation (Web Speech API), high-contrast mode, adjustable text size, and audio read-back — built natively into the application shell, available on every level.
- Design choices were evaluated against WCAG principles and the **EU Accessibility Act (EAA)** compliance trajectory, anticipating regulatory requirements relevant to any multi-region rollout.
- Multimodal redundancy (text, audio, video, interactive) ensures no single representation channel is a single point of failure for any learner profile.

7 CONTENT STRATEGY, MODULARITY & REUSE

- Each level is built as a **self-contained module** (Instruction → Mission → Checkpoint) — enabling reuse or resequencing of individual levels for shorter-format derivative courses without rebuilding the full journey.
- The underlying Swales/Hyland rhetorical-structure engine (dynamic dropdown menus for hedges, boosters, logical markers) is decoupled from the gamification layer, allowing the core linguistic scaffolding to be reused in non-gamified contexts if a future audience requires it.

8 RISKS, DEPENDENCIES & HANDOFF CONSIDERATIONS

Risk / Dependency	Mitigation
Engagement drop-off in fully asynchronous format with no instructor cadence	XP economy + narrative identity arc + boss-level gating to sustain intrinsic motivation without synchronous touchpoints
Reliance on third-party AI tutor (custom GPT) availability	AI tutor positioned as supplementary scaffolding, not a single point of failure for core content delivery
Formal WCAG audit not yet completed at time of launch	Captured explicitly as next-iteration backlog item following independent evaluator recommendation

9 REFERENCE

Eslava-Graterol, A. (2026). Academic Writing Journey [Gamified eLearning unit]. Universitat Politècnica de València. profeanaeslavaedtech.github.io/Gamification/